AI Playbook and Toolkit for Public Health Departments

Introduction to AI Governance for Public Health

GOV 100

AI governance is the set of people, policies, processes, decision rights, documentation practices, and oversight activities that help a public health agency use AI responsibly. Governance does not mean that every staff member must become an AI expert, and it does not mean that every AI idea must be blocked by a committee. Good governance helps an agency decide what AI uses are appropriate, what review is needed, what data may be used, who is accountable, and how staff should respond when something goes wrong. This introductory module is designed for all public health staff, including program staff, technical staff, supervisors, epidemiologists, informaticians, communications staff, policy staff, analysts, and operational leaders. Public health AI governance affects daily work because AI may be used to draft messages, summarize case information, triage service requests, review grants, analyze surveillance signals, support dashboards, write code, translate materials, prioritize outreach, or assist with procurement and contract monitoring. Staff do not need to own the governance process to participate in it. They need to know how to recognize AI use, follow agency rules, document important decisions, protect data, escalate concerns, and keep humans accountable for public health judgment. The module introduces the basic purpose of AI governance, the difference between governance and project management, the roles that commonly participate in governance, and the minimum questions every AI use should answer before it moves forward. Learners will practice applying governance thinking to a realistic public health workflow and will produce a simple governance intake and escalation note that can be adapted for their agency.

Level	introductory
Primary track	Governance and Security Track
Appears in tracks	shared-foundational, governance-security

Learning Objectives

- Define AI governance in plain language and explain why it matters for public health practice.
- Distinguish AI governance from AI policy, IT security review, project management, and individual tool use.
- Identify common AI governance roles, including program owner, data steward, privacy lead, security lead, legal counsel, equity reviewer, communications reviewer, technical reviewer, procurement reviewer, executive sponsor, and frontline user.
- Recognize which AI uses should be escalated for formal review because they involve sensitive data, public-facing information, individual or community impact, resource allocation, surveillance action, vendor products, automation, or public trust risks.
- Apply a basic governance lens to a public health AI use case by documenting purpose, data involved, expected benefit, risk, human review, decision rights, and escalation needs.
- Produce an introductory AI governance intake and escalation note for one real or realistic public health workflow.

Module Overview

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The module introduces the basic purpose of AI governance, the difference between governance and project management, the roles that commonly participate in governance, and the minimum questions every AI use should answer before it moves forward. Learners will practice applying governance thinking to a realistic public health workflow and will produce a simple governance intake and escalation note that can be adapted for their agency.

Definitions

- AI governance: AI governance is the organized approach an agency uses to direct, review, document, monitor, and improve AI use. It includes policies, roles, review pathways, decision rights, evidence requirements, approval conditions, monitoring expectations, escalation processes, and retirement decisions.
- Governance body: A governance body is the group or structure responsible for reviewing AI use and making or recommending decisions. It may be a formal committee, an existing data governance group with expanded responsibilities, a review board, or a coordinated process across leadership, program, IT, privacy, legal, equity, procurement, communications, and technical reviewers.
- Decision rights: Decision rights define who is allowed to approve, reject, pause, revise, monitor, or retire an AI use. Decision rights should be explicit because AI decisions often cross program, technology, legal, privacy, procurement, and public accountability boundaries.

- AI use case: An AI use case describes a specific way AI will be used to support a public health task or workflow. A strong use case identifies the public health purpose, affected users or populations, data involved, AI function, expected benefit, human review process, risks, and measures of success.
- Risk tier: A risk tier is a category used to determine the level of review required. A low-risk internal drafting aid may not need the same review as a tool that prioritizes field response, summarizes case narratives, identifies high-risk communities, or generates public-facing recommendations.
- Human accountability: Human accountability means that a person or accountable role remains responsible for reviewing AI outputs, interpreting context, making decisions, documenting rationale, and responding to errors. AI should not become the invisible decision-maker in a public health workflow.
- Oversight: Oversight is the ongoing review of AI use after approval. It includes monitoring performance, reviewing incidents, checking equity and usability, confirming that approved conditions are followed, and deciding whether the AI use should continue, change, pause, or be retired.
- Escalation: Escalation is the process for raising a question, concern, incident, or proposed high-risk use to the appropriate reviewer or governance body. Staff should know when and how to escalate privacy concerns, incorrect outputs, potential bias, unsafe recommendations, public communication risks, vendor issues, or unapproved use.

Jurisdiction and Agency Policy Note

This module provides national-level training guidance for public health agencies and partners. It is not legal advice and does not replace review by agency legal counsel, privacy officers, procurement officials, civil rights staff, records officers, information security officials, or other authorized decision-makers. Requirements may vary by state, locality, territory, Tribe, agency, funding source, data-sharing agreement, emergency authority, procurement rule, labor agreement, and organizational policy. Learners should use this module to identify the issues that require review and should confirm applicable requirements before approving, procuring, deploying, or publicly communicating about AI-supported work.

Why This Topic Matters for Public Health AI

Public health agencies make decisions that affect people, communities, partners, regulated entities, funding recipients, and the public record. AI can help staff work faster and identify patterns, but it can also introduce new risks. An AI tool may produce a convincing but incorrect summary, expose protected or sensitive information, make uneven recommendations across populations, rely on outdated data, automate a step that should require human judgment, or create confusion about who is responsible for a decision. Governance creates a way to identify these risks before they become routine practice.

Governance is especially important because AI use can start informally. A staff member may paste text into a public chatbot to shorten a report. A program may test a vendor tool before procurement has reviewed data terms. A dashboard team may add an AI-generated interpretation to a public-facing display. An analyst may use code assistance to modify a production script. Each action may seem small, but the combined effect can create privacy, equity, security, quality, procurement, records, and public trust risks if the agency does not have shared expectations.

AI governance also supports innovation. Clear rules help staff know which uses are allowed, which uses require review, and which uses are prohibited. Governance can prevent unnecessary delays by creating a risk-tiered pathway. Low-risk internal productivity uses may need training and supervisor awareness, while higher-risk uses may need privacy, security, equity, legal, data quality, procurement, and communications review. Staff can move faster when they know the pathway.

For public health, governance must connect technology decisions to mission and authority. A technically impressive tool is not enough. The agency must ask whether the tool serves a legitimate public health purpose, uses appropriate data, supports equitable outcomes, preserves human accountability, can be explained to affected communities, and can be monitored over time.

Public Health Example

A local health department wants to use a generative AI tool to help staff summarize long community listening session notes. The goal is to identify common themes related to access to mobile vaccination clinics, language needs, transportation barriers, and trust concerns. Program staff see the tool as a way to save time. Communications staff want to use the themes in a public report. The data team is concerned that some notes include names, small community identifiers, and sensitive comments about immigration status, disability, and prior negative experiences with government services.

Without governance, the project could move forward as a convenience tool. Staff might upload raw notes into an unapproved system, accept the summary without checking whether smaller communities were misrepresented, and publish language that makes the process sound more objective than it is. The agency could unintentionally expose sensitive information, overlook minority viewpoints, or make decisions based on a summary that does not reflect the source material.

With governance, the team would complete a simple use case intake before using the tool. The intake would document the public health purpose, the type of data involved, whether the data include sensitive or identifiable information, who will review the output, how the summary will be validated, whether community voices could be flattened or misrepresented, and whether the output will be used internally or publicly. The governance pathway might require removing identifiers, using an approved secure tool, keeping human review, documenting limitations, and having communications and equity reviewers examine any public-facing summary before release.

The example shows that governance is not only for technical teams. Program staff define the purpose and workflow. Data and privacy staff assess what may be entered into a tool. Equity and community engagement staff consider whether the process preserves community meaning. Communications staff ensure public statements are clear and not overstated. Leadership decides whether the use is appropriate and whether the agency can explain it to the public.

Technical, Programmatic, and Operational Deep Dive

AI governance should be understandable to all staff because the earliest governance decisions often happen before a formal review meeting. The first decision is recognizing that an AI use exists. Staff should treat AI use as more than “using a tool.” If a tool classifies, summarizes, predicts, prioritizes, recommends, generates, translates, extracts, searches, or automates information, the agency should consider whether it falls within AI governance expectations.

The second decision is identifying the purpose. A governance intake should begin with the public health problem, not with the tool. A strong purpose statement explains what workflow problem exists, who is affected, why the current process is insufficient, and what improvement would be meaningful. “Use AI to improve efficiency” is too vague. “Use an approved tool to draft first-pass plain-language versions of internal preparedness messages for human review” is clearer and easier to govern.

The third decision is understanding the data. Public health data may include protected health information, personally identifiable information, small cell counts, tribal data, school data, social service data, behavioral health data, geospatial data, community feedback, images, laboratory data, case notes, or operational records. Governance asks what data will be entered into the system, where it will go, who can access it, whether it can be retained or reused, and what agreements or policies apply.

The fourth decision is defining the role of the AI output. An AI output may be a draft, a suggestion, a risk score, a classification, a summary, a translation, a code recommendation, a dashboard interpretation, or a proposed action. The governance question is whether the output informs human work or changes a decision. A draft that a trained staff member reviews may be low risk. A score that determines who receives outreach first, which complaint receives inspection, or what information appears on a public dashboard needs stronger review.

The fifth decision is assigning human review. Human review must be meaningful. A reviewer should know what to check, what sources to compare against, what errors are most likely, when to reject the output, and how to document the review. A checkbox that says “human reviewed” is not enough if the reviewer lacks the time, authority, or expertise to challenge the output.

The sixth decision is determining the review pathway. A practical governance model uses risk tiers. Low-risk uses may require staff training, approved tools, and supervisor awareness. Moderate-risk uses may require a written use case intake, privacy review, security review, and program approval. High-risk uses may require formal governance review, legal and civil rights input, validation evidence, community engagement, monitoring plans, and leadership approval.

The seventh decision is planning oversight after approval. AI use should not be approved once and forgotten. The agency should know what evidence will be reviewed over time, such as error reports, user feedback, performance measures, equity indicators, security events, public complaints, vendor changes, prompt changes, model changes, workflow changes, and whether the use still serves the original public health purpose.

Risks, Failure Modes, and Guardrails

Unapproved tool use. Staff may use public or personal AI tools before the agency has reviewed privacy, security, retention, or data-use terms. Guardrail: publish a plain-language list of approved, restricted, and prohibited uses, including examples of data that may not be entered into public tools.

Hidden automation. AI may be embedded in vendor products, office software, dashboards, analytics platforms, or workflow tools. Staff may not realize that AI is influencing a recommendation or output. Guardrail: require vendors and project teams to disclose AI functions, data flows, model updates, and human review points.

Overreliance on fluent outputs. Generative AI can produce confident language that appears accurate but is unsupported, incomplete, or wrong. Guardrail: require source verification, human review, and clear labeling of outputs as drafts or decision support when appropriate.

Privacy and confidentiality failures. Sensitive public health information can be exposed through inappropriate prompts, file uploads, model training, vendor retention, access controls, or shared outputs. Guardrail: require data classification, minimum necessary use, approved environments, access controls, and privacy review for sensitive data.

Equity and representation harms. AI can amplify bias in source data, miss smaller populations, flatten community context, or produce recommendations that increase burdens for groups already experiencing inequities. Guardrail: require equity review, subgroup monitoring when appropriate, missing voices analysis, and community-informed interpretation for consequential uses.

Unclear accountability. Staff may not know who approved an AI use, who is responsible for errors, or who can pause the workflow. Guardrail: document owners, decision rights, approval conditions, escalation contacts, and pause/retirement criteria.

Governance overload. Agencies can create review processes so burdensome that staff avoid them or innovation stalls. Guardrail: use risk tiers, standard intake forms, clear timelines, and lightweight pathways for low-risk uses while preserving stronger review for consequential uses.

Application to AI-Supported Workflows

In AI-supported workflows, governance should be visible at the points where work changes. Before use, staff should know whether the tool is approved, what data may be used, what the intended purpose is, and whether the use requires intake or review. During use, staff should know how to review outputs, document decisions, protect sensitive information, and report problems. After use, the agency should be able to audit what was approved, what evidence supported the decision, what outputs were used, and whether the workflow remains safe and useful.

For program staff, governance means connecting AI use to program goals, service delivery, community impact, and workflow reality. Program staff should be able to explain why AI is being considered, who will use it, what decision or task it supports, what could go wrong, and how staff will know whether it is helping.

For technical staff, governance means making the system understandable, secure, testable, and monitorable. Technical staff should document data flows, access controls, model or tool behavior, integration points, logging, validation evidence, change management, and incident response expectations.

For supervisors and leaders, governance means ensuring that staff have clear rules, enough training, a safe way to raise concerns, and a realistic workload for human review. Leaders should avoid approving AI tools without resources for training, monitoring, documentation, and improvement.

For all staff, governance means pausing when an AI use involves sensitive data, public-facing content, decision support, automated prioritization, community impact, legal obligations, or uncertainty about agency policy. In those situations, escalation is not a barrier. It is part of responsible public health practice.

Reflection Questions

Where might AI already be entering your work through drafting tools, analytics tools, vendor platforms, search tools, translation tools, dashboards, code assistance, or workflow automation?

What AI uses in your program area would be low risk, and what uses would require formal review before moving forward?

Who in your agency would need to review an AI use that involves sensitive data, public-facing communication, resource prioritization, or surveillance action?

What information would you need before you could explain an AI-supported workflow to a supervisor, partner, community member, or governance body?

What would make staff comfortable raising concerns about AI outputs or unapproved use without fear of blame?

Practical Exercise

Select one real or realistic public health workflow where AI might be used. Examples include summarizing meeting notes, drafting public health messages, classifying environmental health complaints, reviewing grant applications, supporting case investigation, translating outreach materials, generating code for a dashboard, prioritizing outreach lists, or summarizing surveillance trends.

Complete a one-page introductory AI governance intake and escalation note. The note should include: the public health purpose; the staff or partners who would use the tool; the data involved; whether the data are sensitive or identifiable; the AI function; the expected benefit; who will review the output; what decision or action the output could influence; likely risks; required reviewers; escalation triggers; and whether the use should be low, moderate, or high risk.

Practical exercise example: A chronic disease program wants to use an AI tool to summarize open-ended survey responses from community members about barriers to diabetes prevention classes. The governance note identifies the purpose as reducing manual review burden while preserving community meaning. The data include open-ended comments, ZIP codes, race/ethnicity, age group, and some comments that mention disability, transportation, immigration concerns, and distrust of government. The AI function is summarization and theme extraction. The expected benefit is faster identification of barriers to participation. The risks include privacy, small-community identifiability, loss of minority viewpoints, inaccurate summaries, and overconfidence in themes. The note recommends using only an approved secure tool, removing direct identifiers, keeping the original comments available for human review, checking themes across demographic groups, involving the community engagement lead, and requiring communications review before any public summary is released. The risk tier is moderate because the output may shape program planning and public communication but will not directly determine individual eligibility or services.

After completing the note, identify one governance question that remains unresolved. For example: Can this data be entered into the proposed tool under current data-use agreements? Who must approve public language based on AI-assisted summaries? What threshold would make the project high risk? What evidence would show that the AI summary did not erase smaller community concerns?

Expected Artifact or Evidence

Learners should produce an introductory AI governance intake and escalation note for one public health workflow. The artifact should include a purpose statement, data description, AI function, likely users, expected benefit, risk tier, human review plan, required reviewers, escalation triggers, and at least one unresolved governance question. The artifact can be used as a training record, a draft intake form, or a starting point for a real governance discussion.

Expected Assignment

- Learners should produce an introductory AI governance intake and escalation note for one public health workflow. The artifact should include a purpose statement, data description, AI function, likely users, expected benefit, risk tier, human review plan, required reviewers, escalation triggers, and at least one unresolved governance question. The artifact can be used as a training record, a draft intake form, or a starting point for a real governance discussion.

Knowledge Check

- 1. Which statement best describes AI governance in a public health agency?
- 2. Which AI use should most clearly be escalated for formal review before implementation?
- 3. What is the most important reason to define human review in an AI-supported workflow?
- 4. A staff member wants to paste identifiable case notes into a public AI chatbot to create a quick summary. What should happen first?
- 5. Which item belongs in a basic AI governance intake note?

References and Resources

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